

SKILLS

Generative AI Systems:	Agentic AI, LangGraph, LangChain, LangSmith (Tracing & Evaluation), MCP, RAG
Natural Language Processing (NLP):	Fine-tuning, Prompt Engineering, Hugging Face, Transformers
Machine Learning & Analytics:	Supervised/Unsupervised Learning, Feature Engineering, Model Evaluation
Programming & Data Science:	Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, Jupyter)
Web Development & APIs:	FastAPI, Flask, Django, RESTful APIs, HTML, CSS, JavaScript, Bootstrap, Tailwind
Mobile & Backend Services:	Flutter, Firebase, Node.js (Cloud Functions)
Databases & Big Data:	SQL, PostgreSQL, Query Optimization, Firebase
Data Visualization & BI Tools:	Power BI, Plotly, Dash
DevOps & Tools:	Docker, Git/GitHub, CI/CD Pipelines, Linux/Unix, AWS
Productivity Tools:	MS Excel (Advanced), MS Word, MS PowerPoint, LaTeX

EXPERIENCE

Zalio GmbH, Junior AI Engineer | Munich, Germany

Mar 2026 – Present

- Integrating specialized sub-agent endpoints into the chat interface to enable faster, task-specific AI interactions.
- Designing and developing a multi-agent system for procurement workflows, improving automation and decision support.
- Enhancing prompt engineering strategies to improve response quality, reliability, and task alignment of AI agents.
- Upgrading AI infrastructure using LangChain and LangGraph to improve modularity, scalability, and maintainability.
- Developing frontend components using Next.js with Atomic Design principles for a scalable UI architecture.
- Building backend services using FastAPI and deploying them on AWS EC2 to support robust and scalable AI pipelines.
- Implementing CI/CD pipelines to streamline development, testing, and deployment processes.
- Collaborating on technical documentation and translating business requirements into actionable engineering solutions in a fast-paced startup environment.

Siemens Energy, Master's Thesis – Junior Data Scientist (NLP) | Erlangen, Germany

Jul 2025 – Feb 2026

- Designed and implemented a modular, production-oriented multi-agent framework for LLM-based systems with explicit orchestration, persistent conversation state, and auditable execution flows.
- Implemented agent orchestration using LangGraph with centralized routing, hierarchical agent coordination, and strict separation between orchestration, agent execution, memory management, and tool invocation.
- Built a unified execution pipeline exposed via a FastAPI-based REST interface with OpenAI-compatible endpoints, enabling consistent behavior across CLI, web UI, and API clients with streaming responses.
- Integrated PostgreSQL-based checkpointing for persistent agent memory, reproducibility, and post-hoc auditability of multi-step agent executions.
- Applied the framework to two Siemens Energy use cases: a Factory Assistant for engineering requirements and report generation, and a Terminal Management System Assistant for log analysis, diagnostic summaries, and email drafting.
- Implemented safety-critical human-in-the-loop approvals for external actions such as email sending, ensuring oversight and preventing unintended operations.

Ilham Labs.ai, Founder & Product Engineer | Independent Product Initiative, Germany

2025 – Present

- Founded an independent product initiative and owned product strategy, architecture, full-stack development, deployment, security, and iteration.
- Built Lattice Lab, a visual multi-LLM platform supporting parallel model execution, real-time streaming, branching workflows, context merging, authentication, and PostgreSQL persistence.
- Built whenyouopen.me and deployed containerized production workloads on AWS EC2 and AWS Lightsail using FastAPI, AWS S3, Docker, Nginx, SSL, encryption, and automated deployment workflows.

Institute for Employment Research, Research Assistant | Nuremberg, Germany

Nov 2024 – Feb 2026

- Applied ML techniques (XGBoost, Random Forest, SVM, PU learning: Elkan & Noto, Two-Step Strategy, Bagging) to forecast labor market trends and analyze job suitability patterns.
- Built scalable data pipelines for preprocessing and feature engineering on 90M+ records, optimizing for efficient data processing and management.
- Developed predictive models to rank job offers and aid caseworkers in pre-selecting opportunities, addressing class imbalances through weight estimation to improve model robustness.
- Conducted comprehensive data analysis, visualization, and EDA to extract key labor market insights and deliver findings through reports and stakeholder presentations.
- Contributed to research paper writing, proofreading, and technical documentation for academic publications.

BASF, Data Consultant Intern | Ludwigshafen am Rhein, Germany

Apr 2025 – Jul 2025

- Supported internal clients across business units by building and integrating Generative AI tools into daily workflows.
- Developed a Data Validation AI Agent to analyze structured datasets, generate validation reports, compute quality scores, and run custom validation tests.
- Led the end-to-end event app for Inhouse Consulting (submissions, voting, UI, analytics and access controls).
- Contributed to internal project management tools to streamline team operations and task tracking.

Shape AI, Business Development Intern | Delhi, India

Mar 2021 – Oct 2021

- Performed data analysis using Power BI, extracting actionable insights from sales data to drive business decisions.
- Developed dashboards and visualizations to clearly communicate key metrics across the organization.
- Designed a data-backed course structure for the winter training program, boosting engagement and enrollment.
- Contributed to a marketing project by analyzing target audience behavior, improving course promotion strategies.

- Built responsive websites using HTML, CSS, Bootstrap, and JavaScript, ensuring performance and cross-browser support.
- Performed testing and debugging on various devices, improving user experience and accessibility.
- Optimized JavaScript and CSS for faster load times, boosting website interactivity and performance.
- Adopted modern web frameworks and best practices, driving continuous improvements in development processes.

PROJECTS

Multi Agent Project Generator - Autonomous AI Project Builder

- Developed an intelligent multi-agent system using LangGraph and Google Gemini AI that automatically generates complete, production-ready projects from natural language prompts.
- Implemented a Planner–Architect–Coder agent hierarchy for structured planning, architecture design, and code generation with Pydantic-based communication.
- Secured project creation and validation through modular tasks and sandboxed file operations.
- Supports multiple tech stacks with configurable recursion and type-safe structured outputs.

URL: github.com/rebel47/Multi-Agent-Project-Generator

Ilmora – Private On-Device LLM Inference Platform (Flutter)

- Built a production-ready on-device LLM app with Flutter, enabling fully offline, privacy-first inference.
- Implemented intelligent inference workflows: coherence checks, temperature decay retries, token-budgeted prompts.
- Implemented inference workflows: coherence checks, temperature decay retries, token-budgeted prompt construction.
- Designed persistent multi-session memory with local storage, real-time search, and JSON-based export/import.
- Optimized native inference via llama.cpp bindings, Android NDK, CMake, with quantization detection and UI badges.

URL: Website: Ilmora

Fine-Tuned GPT-2 for Poetry Generation with RL

- Fine-tuned GPT-2 (124M) on a Gutenberg poetry corpus, leveraging Reinforcement Learning (RL) to boost stylistic quality and coherence.
- Designed custom reward functions for rhyme, coherence, and creativity.
- Used Transformers for training with adaptive learning rates and policy-gradient methods.
- Published on the Hugging Face Hub; 3600+ downloads with public inference.

URL: [ayazfau/GPT2-124M-poetry-RL](https://huggingface.co/ayazfau/GPT2-124M-poetry-RL)

CERTIFICATES

Huggingface - AI Agents Fundamentals - Agents fundamentals, Tools and Action, Agents Workflow

iNeuron.ai - Full Stack Data Analytics - Python, SQL, Snowflake, PowerBI, DAX, Excel

iNeuron.ai - Deep Learning Foundation - Neural Network, Backpropagation, CNN

Kaggle - Intro to Machine Learning - Model Validation, Underfitting-Overfitting, Random Forest

freeCodeCamp - Data Analysis with Python - Numpy, Pandas, Data Cleaning

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg, MSc. Data Science | Erlangen, Germany

Oct 2022 - Present

• Courses: Business Analytics, ML Time Series, ML in Finance, Mathematics of Learning, Linked Data

Guru Gobind Singh Indraprastha University, Bachelors of Computer Applications | Delhi, India

Aug 2018 - Sept 2021

• Courses: OOPs, Network Security, DBMS, Web Technologies, Data Warehousing and Data Mining

Languages

English Professional proficiency

German A2 proficiency

Hindi & Urdu Native proficiency